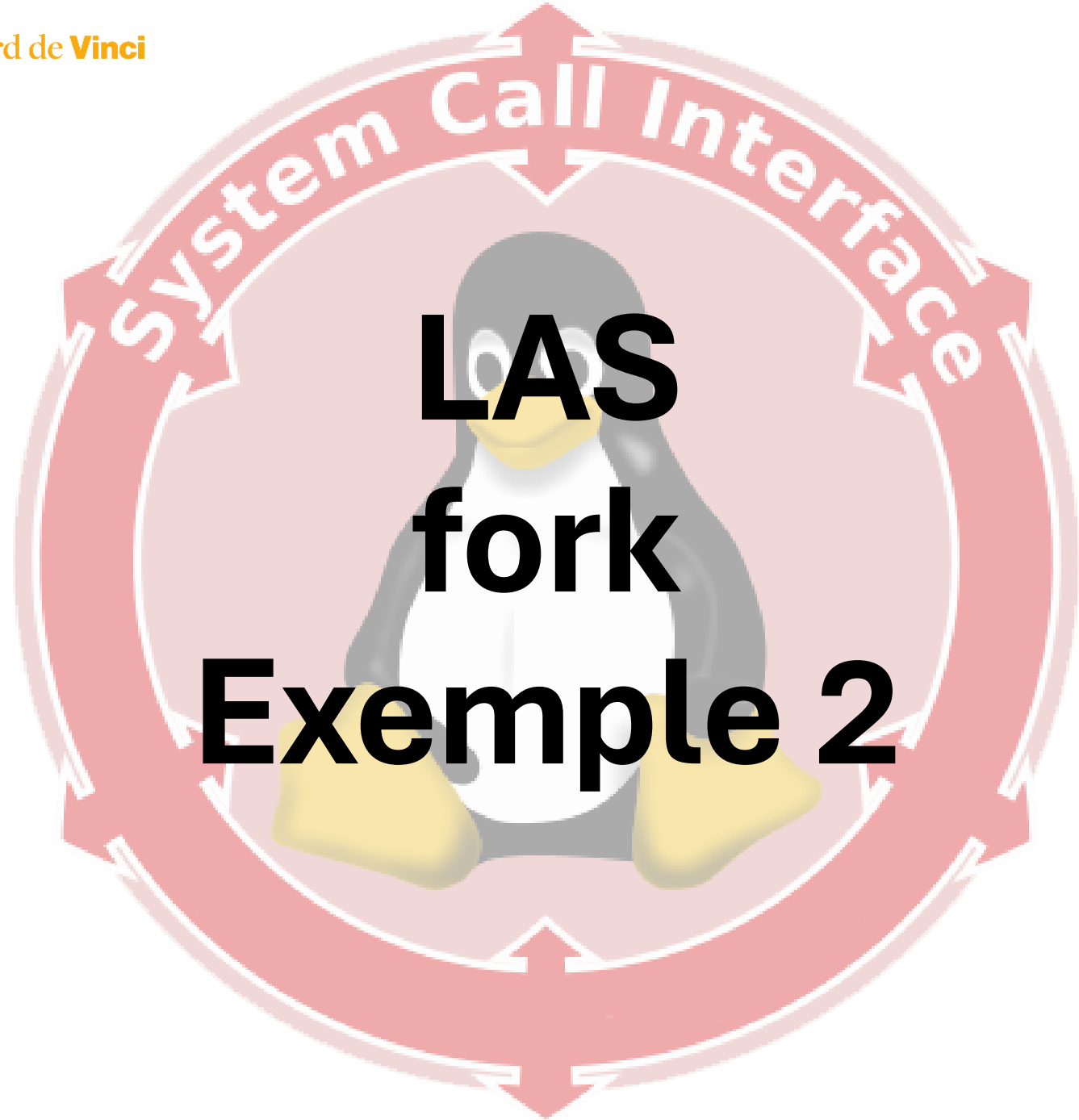
Haute École Léonard de Vinci
SCIENCES ET TECHNIQUES




# LAS
# fork
# Exemple 2

```
alegrand@courslinux:LAS_02_exemples$ lastrace.sh exemple2
Processus parent (pid=2993688) en attente de la fin de son enfant (pid fils=2993689)
Je suis le fils './exemple2': pid=2993689 - ppid=2993688
F S   UID     PID    PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
0 S  1002 2991417 2991416  0  80   0 -  2503 do_wai pts/0    00:00:01 bash
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0 S  1002 2993688 2993685  0  80   0 -   617 do_wai pts/0    00:00:00 exemple2
0 R  1002 2993689 2993688 99  80   0 -  2771 -      pts/0    00:00:00 ps
Processus parent se termine après son enfant (pid fils=2993689)

Traced syscalls and signals on ./exemple2:

[PID=2993688] execve("./exemple2", ["./exemple2"], 0x7ffc22d10a48 /* 23 vars */) = 0
[PID=2993688] openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
[PID=2993688] close(3) = 0
[PID=2993688] openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
[PID=2993688] read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\20t\2\0\0\0\0\0"..., 832) = 832
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[PID=2993688] clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, child_tidptr=
0x7f6d66086a10) = 2993689
[PID=2993688] write(1, "\33[1;35mProcessus parent (pid=299"..., 92) = 92
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[PID=2993689] write(1, "\33[1;33mJe suis le fils './exempl"..., 64) = 64
[PID=2993689] execve("/usr/bin/ps", ["ps", "-l"], 0x7ffee6b63308 /* 23 vars */) = 0
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Exécution de la commande `ps`...

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[PID=2993689] write(1, "0 S 1002 2991417 2991416 0 80"..., 75) = 75
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[PID=2993688] write(1, "\33[1;35mProcessus parent se termi"..., 72) = 72
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**Lancement de ./exemple02 (PID = 2993688) par le script *lastrace.sh***

Exécution de la commande `ps`...

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**PÈRE:**
**création du processus fils → *PID = 2993689***

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**PÈRE:**
**écriture d'un message sur stdout**

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**PÈRE:**
**le processus père se met en attente de la terminaison de son fils**

Exécution de la commande `ps`...

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**FILS:**
**écriture d'un message sur stdout**

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[PID=2993689] close(2) = 0
[PID=2993689] +++ exited with 0 +++
[PID=2993688] <... wait4 resumed>NULL, 0, NULL) = 2993689
[PID=2993688] --- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=2993689, si_uid=1002, si_status=0,
 si_utime=2 /* 0.02 s */, si_stime=22 /* 0.22 s */} ---
[PID=2993688] write(1, "\33[1;35mProcessus parent se termi"..., 72) = 72
[PID=2993688] +++ exited with 0 +++
```

**FILS: remplacement du processus par la commande ps**

```
alegrand@courslinux:LAS_02_exemples$ lastrace.sh exemple2
Processus parent (pid=2993688) en attente de la fin de son enfant (pid fils=2993689)
Je suis le fils './exemple2': pid=2993689 - ppid=2993688
F S   UID     PID    PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
0 S  1002 2991417 2991416  0  80   0 -  2503 do_wai pts/0    00:00:01 bash
0 S  1002 2993684 2991417  0  80   0 -  1733 do_wai pts/0    00:00:00 lastrace.sh
0 S  1002 2993685 2993684 99  80   0 -  2003 do_wai pts/0    00:00:00 strace
0 S  1002 2993688 2993685  0  80   0 -   617 do_wai pts/0    00:00:00 exemple2
0 R  1002 2993689 2993688 99  80   0 -  2771 -      pts/0    00:00:00 ps
Processus parent se termine après son enfant (pid fils=2993689)

Traced syscalls and signals on ./exemple2:

[PID=2993688] execve("./exemple2", ["./exemple2"], 0x7ffc22d10a48 /* 23 vars */) = 0
[PID=2993688] openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
[PID=2993688] close(3) = 0
[PID=2993688] openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
[PID=2993688] read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\20t\2\0\0\0\0\0"..., 832) = 832
[PID=2993688] close(3) = 0
[PID=2993688] clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, child_tidptr=
0x7f6d66086a10) = 2993689
[PID=2993688] write(1, "\33[1;35mProcessus parent (pid=299"..., 92) = 92
[PID=2993688] wait4(-1, <unfinished ...>
[PID=2993689] write(1, "\33[1;33mJe suis le fils './exempl"..., 64) = 64
[PID=2993689] execve("/usr/bin/ps", ["ps", "-l"], 0x7ffee6b63308 /* 23 vars */) = 0
```

Exécution de la commande `ps`...

```
[PID=2993689] write(1, "0 S 1002 2991417 2991416 0 80"..., 75) = 75
[PID=2993689] write(1, "0 S 1002 2993684 2991417 0 80"..., 82) = 82
[PID=2993689] write(1, "0 S 1002 2993685 2993684 99 80"..., 77) = 77
[PID=2993689] write(1, "0 S 1002 2993688 2993685 0 80"..., 79) = 79
[PID=2993689] write(1, "0 R 1002 2993689 2993688 99 80"..., 73) = 73
[PID=2993689] close(1) = 0
[PID=2993689] close(2) = 0
[PID=2993689] +++ exited with 0 +++
[PID=2993688] <... wait4 resumed>NULL, 0, NULL) = 2993689
[PID=2993688] --- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=2993689, si_uid=1002, si_status=0,
 si_utime=2 /* 0.02 s */, si_stime=22 /* 0.22 s */} ---
[PID=2993688] write(1, "\33[1;35mProcessus parent se termi"..., 72) = 72
[PID=2993688] +++ exited with 0 +++
```

**FILS:**
**écriture de l'output de la commande ps sur stdout**

```
alegrand@courslinux:LAS_02_exemples$ lastrace.sh exemple2
Processus parent (pid=2993688) en attente de la fin de son enfant (pid fils=2993689)
Je suis le fils './exemple2': pid=2993689 - ppid=2993688
F S   UID     PID    PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
0 S  1002 2991417 2991416  0  80   0 -  2503 do_wai pts/0    00:00:01 bash
0 S  1002 2993684 2991417  0  80   0 -  1733 do_wai pts/0    00:00:00 lastrace.sh
0 S  1002 2993685 2993684 99  80   0 -  2003 do_wai pts/0    00:00:00 strace
0 S  1002 2993688 2993685  0  80   0 -   617 do_wai pts/0    00:00:00 exemple2
0 R  1002 2993689 2993688 99  80   0 -  2771 -      pts/0    00:00:00 ps
Processus parent se termine après son enfant (pid fils=2993689)

Traced syscalls and signals on ./exemple2:

[PID=2993688] execve("./exemple2", ["./exemple2"], 0x7ffc22d10a48 /* 23 vars */) = 0
[PID=2993688] openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
[PID=2993688] close(3) = 0
[PID=2993688] openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
[PID=2993688] read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\20t\2\0\0\0\0\0"..., 832) = 832
[PID=2993688] close(3) = 0
[PID=2993688] clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, child_tidptr=
0x7f6d66086a10) = 2993689
[PID=2993688] write(1, "\33[1;35mProcessus parent (pid=299"..., 92) = 92
[PID=2993688] wait4(-1, <unfinished ...>
[PID=2993689] write(1, "\33[1;33mJe suis le fils './exempl"..., 64) = 64
[PID=2993689] execve("/usr/bin/ps", ["ps", "-l"], 0x7ffee6b63308 /* 23 vars */) = 0
```

Exécution de la commande `ps`...

```
[PID=2993689] write(1, "0 S 1002 2991417 2991416 0 80"..., 75) = 75
[PID=2993689] write(1, "0 S 1002 2993684 2991417 0 80"..., 82) = 82
[PID=2993689] write(1, "0 S 1002 2993685 2993684 99 80"..., 77) = 77
[PID=2993689] write(1, "0 S 1002 2993688 2993685 0 80"..., 79) = 79
[PID=2993689] write(1, "0 R 1002 2993689 2993688 99 80"..., 73) = 73
[PID=2993689] close(1) = 0
[PID=2993689] close(2) = 0
[PID=2993689] +++ exited with 0 +++
[PID=2993688] <... wait4 resumed>NULL, 0, NULL) = 2993689
[PID=2993688] --- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=2993689, si_uid=1002, si_status=0,
 si_utime=2 /* 0.02 s */, si_stime=22 /* 0.22 s */} ---
[PID=2993688] write(1, "\33[1;35mProcessus parent se termi"..., 72) = 72
[PID=2993688] +++ exited with 0 +++
```

**FILS:**
**fin de l'exécution du fils**
**(PID = 2993689)**
**avec exit code = 0**

```
alegrand@courslinux:LAS_02_exemples$ lastrace.sh exemple2
Processus parent (pid=2993688) en attente de la fin de son enfant (pid fils=2993689)
Je suis le fils './exemple2': pid=2993689 - ppid=2993688
F S   UID     PID    PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
0 S  1002 2991417 2991416  0  80   0 -  2503 do_wai pts/0    00:00:01 bash
0 S  1002 2993684 2991417  0  80   0 -  1733 do_wai pts/0    00:00:00 lastrace.sh
0 S  1002 2993685 2993684 99  80   0 -  2003 do_wai pts/0    00:00:00 strace
0 S  1002 2993688 2993685  0  80   0 -   617 do_wai pts/0    00:00:00 exemple2
0 R  1002 2993689 2993688 99  80   0 -  2771 -      pts/0    00:00:00 ps
Processus parent se termine après son enfant (pid fils=2993689)

Traced syscalls and signals on ./exemple2:

[PID=2993688] execve("./exemple2", ["./exemple2"], 0x7ffc22d10a48 /* 23 vars */) = 0
[PID=2993688] openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
[PID=2993688] close(3) = 0
[PID=2993688] openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
[PID=2993688] read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\20t\2\0\0\0\0\0"..., 832) = 832
[PID=2993688] close(3) = 0
[PID=2993688] clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, child_tidptr=
0x7f6d66086a10) = 2993689
[PID=2993688] write(1, "\33[1;35mProcessus parent (pid=299"..., 92) = 92
[PID=2993688] wait4(-1, <unfinished ...>
[PID=2993689] write(1, "\33[1;33mJe suis le fils './exempl"..., 64) = 64
[PID=2993689] execve("/usr/bin/ps", ["ps", "-l"], 0x7ffee6b63308 /* 23 vars */) = 0
```

Exécution de la commande `ps`...

```
[PID=2993689] write(1, "0 S 1002 2991417 2991416 0 80"..., 75) = 75
[PID=2993689] write(1, "0 S 1002 2993684 2991417 0 80"..., 82) = 82
[PID=2993689] write(1, "0 S 1002 2993685 2993684 99 80"..., 77) = 77
[PID=2993689] write(1, "0 S 1002 2993688 2993685 0 80"..., 79) = 79
[PID=2993689] write(1, "0 R 1002 2993689 2993688 99 80"..., 73) = 73
[PID=2993689] close(1) = 0
[PID=2993689] close(2) = 0
[PID=2993689] +++ exited with 0 +++
[PID=2993688] <... wait4 resumed>NULL, 0, NULL) = 2993689
[PID=2993688] --- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=2993689, si_uid=1002, si_status=0,
 si_utime=2 /* 0.02 s */, si_stime=22 /* 0.22 s */} ---
[PID=2993688] write(1, "\33[1;35mProcessus parent se termi"..., 72) = 72
[PID=2993688] +++ exited with 0 +++
```

**PÈRE:**
**le processus père sort de son attente et récupère le pid du fils qui s'est terminé (PID = 2993689)**

```
alegrand@courslinux:LAS_02_exemples$ lastrace.sh exemple2
Processus parent (pid=2993688) en attente de la fin de son enfant (pid fils=2993689)
Je suis le fils './exemple2': pid=2993689 - ppid=2993688
F S   UID     PID    PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
0 S  1002 2991417 2991416  0  80   0 -  2503 do_wai pts/0    00:00:01 bash
0 S  1002 2993684 2991417  0  80   0 -  1733 do_wai pts/0    00:00:00 lastrace.sh
0 S  1002 2993685 2993684 99  80   0 -  2003 do_wai pts/0    00:00:00 strace
0 S  1002 2993688 2993685  0  80   0 -   617 do_wai pts/0    00:00:00 exemple2
0 R  1002 2993689 2993688 99  80   0 -  2771 -      pts/0    00:00:00 ps
Processus parent se termine après son enfant (pid fils=2993689)

Traced syscalls and signals on ./exemple2:

[PID=2993688] execve("./exemple2", ["./exemple2"], 0x7ffc22d10a48 /* 23 vars */) = 0
[PID=2993688] openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
[PID=2993688] close(3) = 0
[PID=2993688] openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
[PID=2993688] read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\20t\2\0\0\0\0\0"..., 832) = 832
[PID=2993688] close(3) = 0
[PID=2993688] clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, child_tidptr=
0x7f6d66086a10) = 2993689
[PID=2993688] write(1, "\33[1;35mProcessus parent (pid=299"..., 92) = 92
[PID=2993688] wait4(-1, <unfinished ...>
[PID=2993689] write(1, "\33[1;33mJe suis le fils './exempl"..., 64) = 64
[PID=2993689] execve("/usr/bin/ps", ["ps", "-l"], 0x7ffee6b63308 /* 23 vars */) = 0
```

Exécution de la commande `ps`...

```
[PID=2993689] write(1, "0 S 1002 2991417 2991416 0 80"..., 75) = 75
[PID=2993689] write(1, "0 S 1002 2993684 2991417 0 80"..., 82) = 82
[PID=2993689] write(1, "0 S 1002 2993685 2993684 99 80"..., 77) = 77
[PID=2993689] write(1, "0 S 1002 2993688 2993685 0 80"..., 79) = 79
[PID=2993689] write(1, "0 R 1002 2993689 2993688 99 80"..., 73) = 73
[PID=2993689] close(1) = 0
[PID=2993689] close(2) = 0
[PID=2993689] +++ exited with 0 +++
[PID=2993688] <... wait4 resumed>NULL, 0, NULL) = 2993689
[PID=2993688] --- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=2993689, si_uid=1002, si_status=0,
 si_utime=2 /* 0.02 s */, si_stime=22 /* 0.22 s */} ---
[PID=2993688] write(1, "\33[1;35mProcessus parent se termi"..., 72) = 72
[PID=2993688] +++ exited with 0 +++
```

**PÈRE:**
**le père affiche un message sur stdout**

```
alegrand@courslinux:LAS_02_exemples$ lastrace.sh exemple2
Processus parent (pid=2993688) en attente de la fin de son enfant (pid fils=2993689)
Je suis le fils './exemple2': pid=2993689 - ppid=2993688
F S   UID     PID    PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
0 S  1002 2991417 2991416  0  80   0 -  2503 do_wai pts/0    00:00:01 bash
0 S  1002 2993684 2991417  0  80   0 -  1733 do_wai pts/0    00:00:00 lastrace.sh
0 S  1002 2993685 2993684 99  80   0 -  2003 do_wai pts/0    00:00:00 strace
0 S  1002 2993688 2993685  0  80   0 -   617 do_wai pts/0    00:00:00 exemple2
0 R  1002 2993689 2993688 99  80   0 -  2771 -      pts/0    00:00:00 ps
Processus parent se termine après son enfant (pid fils=2993689)

Traced syscalls and signals on ./exemple2:

[PID=2993688] execve("./exemple2", ["./exemple2"], 0x7ffc22d10a48 /* 23 vars */) = 0
[PID=2993688] openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
[PID=2993688] close(3) = 0
[PID=2993688] openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
[PID=2993688] read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\20t\2\0\0\0\0\0"..., 832) = 832
[PID=2993688] close(3) = 0
[PID=2993688] clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, child_tidptr=
0x7f6d66086a10) = 2993689
[PID=2993688] write(1, "\33[1;35mProcessus parent (pid=299"..., 92) = 92
[PID=2993688] wait4(-1, <unfinished ...>
[PID=2993689] write(1, "\33[1;33mJe suis le fils './exempl"..., 64) = 64
[PID=2993689] execve("/usr/bin/ps", ["ps", "-l"], 0x7ffee6b63308 /* 23 vars */) = 0
```

Exécution de la commande `ps`...

```
[PID=2993689] write(1, "0 S 1002 2991417 2991416 0 80"..., 75) = 75
[PID=2993689] write(1, "0 S 1002 2993684 2991417 0 80"..., 82) = 82
[PID=2993689] write(1, "0 S 1002 2993685 2993684 99 80"..., 77) = 77
[PID=2993689] write(1, "0 S 1002 2993688 2993685 0 80"..., 79) = 79
[PID=2993689] write(1, "0 R 1002 2993689 2993688 99 80"..., 73) = 73
[PID=2993689] close(1) = 0
[PID=2993689] close(2) = 0
[PID=2993689] +++ exited with 0 +++
[PID=2993688] <... wait4 resumed>NULL, 0, NULL) = 2993689
[PID=2993688] --- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=2993689, si_uid=1002, si_status=0,
 si_utime=2 /* 0.02 s */, si_stime=22 /* 0.22 s */} ---
[PID=2993688] write(1, "\33[1;35mProcessus parent se termi"..., 72) = 72
[PID=2993688] +++ exited with 0 +++
```

**PÈRE:**
**fin de l'exécution du père (PID = 2993689) avec exit code = 0**